

# Parametric Multimodal User Memory: Storing What Captions Cannot Carry

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## Abstract

A personalized AI agent needs a *user memory*: a persistent model of who its user is, accumulated across many encounters and consulted on each new one. Today that memory is almost always stored as *text*. The agent keeps transcripts of what was said and captions of what was seen, then retrieves them by similarity search over those words. This works for the *captionable* slice of a person (“my cat is named Bibi”), but it silently drops a second slice that no caption can hold. That second slice is the *perceptual* one: how a user’s voice sounds, how their face reads across lighting and age, how tired they sound today against their own baseline. The moment such content is written down, the signal that tells one person from another is gone. We argue that perceptual user memory needs a different primitive, one that keeps the content in its native modality instead of projecting it through text.

We introduce a **training-free, in-model multimodal user memory**. The agent stores each perception directly, as a row in a small per-modality bank attached to a frozen language model, pairing the perception’s encoder embedding (the key) with the model’s own vector for a marker token (the value), and recalls it by attention at the output head, with no captioning step in between. Registering a new identity is a single tensor append. Recall takes constant time no matter how many are stored. The mechanism needs *no training*: with a sharp read it reproduces the encoder’s own recall *exactly*, and it does so on every frozen model we tried (ten families from 1.5B to 14B, tied and untied embeddings, transformer and hybrid-Mamba). It does not beat the encoder’s recall, which is the encoder’s job. What it adds is that the recalled identity becomes a *token the model conditions on*, so the perception lives inside generation and composes with text memory, which an external retrieval index cannot do. On **PerceptMem**, a new benchmark spanning faces, voices, painting style, room acoustics, and tone of voice, we map what such a memory can and cannot hold: perceptual identity is capacity-limited and degrades gracefully ( $\text{recall} \approx \min(1, k/M)$  per slot), while exact facts are binding-limited and belong in a text store. This memory sits *alongside* text memory rather than replacing it, and ordinary text recall is preserved exactly. An agent with both remembers what its user said *and* what they were like.

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Code: <https://github.com/19PINE-AI/multimodal-user-memory>

## 1 Introduction

When an AI agent “remembers” something about the person it serves, what does it keep? In almost every system today, *text*. Mem0 [Chhikara et al., 2025], MemoryLLM [Wang et al., 2024],

MemGPT / Letta [Packer et al., 2023, Team, 2024], M3-Agent [Long et al., 2025], and the wider conversational-memory line [Wu et al., 2024] turn what the agent saw or heard into transcripts (via ASR [Radford et al., 2022]) and captions (via BLIP-2 [Li et al., 2023] or LLaVA [Liu et al., 2023]), store the fragments in a sentence-encoder vector index [Reimers and Gurevych, 2019], and search them later. Personalized vision-language systems (MyVLM [Alaluf et al., 2024], Yo’LLaVA [Nguyen et al., 2024], MC-LLaVA [An et al., 2024], Online-PVLM [Bai et al., 2025], RAP [Hao et al., 2025]) keep the same textual spine: a *named* concept (“Bibi”) is the handle, the perception an afterthought.

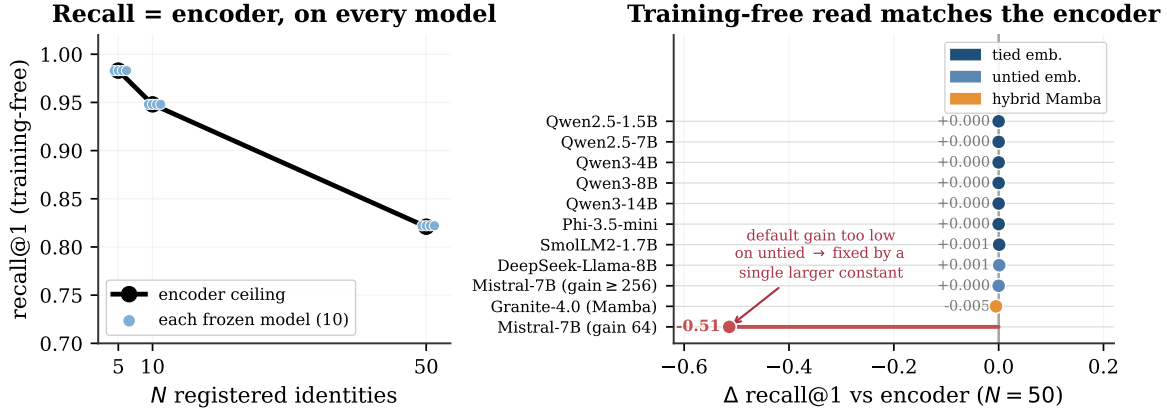
This is exactly right for one half of a person. “My cat is named Bibi.” “My favourite restaurant is in Paris.” “I am vegetarian.” These are *captionable*: they pass through text intact and come back whole. But they are not the whole person. Consider what else an attentive companion carries about someone:

- How their voice sounds when they say their own name, and how that differs from a stranger saying it.
- How their face reads today: a little older than last year’s photo, but unmistakably them.
- How their brushwork looks: “this is by the painter from Tuesday, even though it’s from a different period.”
- What their kitchen sounds like: “you’re calling from the same room as yesterday.”
- How tired they sound today, relative to their baseline last week.

None of this is captionable. “A brown-haired man” does not tell two brown-haired men apart; “the user sounded tired” cannot separate today’s tired from last week’s. Writing these down throws away the very signal that made them discriminative, and no better caption closes the gap: it is a property of the channel. **Text is a lossy medium for perceptual content, and the loss is fundamental to the modality.**

So a personalized agent’s memory has two halves. *Captionable* content (facts, preferences, propositions) belongs in a text store, where sentence-encoder similarity works beautifully. *Perceptual* content — how a face, a voice, a room, or a painting actually *is* — needs a primitive that keeps it in its native modality, and today it has no home. It is either flattened into a caption (and lost) or handed to a recognition tool that returns a bare label like “speaker 47”, so the perception never reaches the model. The missing piece is a memory that stores a perception *as a perception*.

**Our proposal.** We introduce **parametric multimodal user memory**: instead of captioning a perception, the agent stores it directly, as a row in a small per-modality bank bolted onto a frozen language model. Each row pairs the raw encoder embedding of the perception (the key) with the model’s own internal vector for a marker token (the value). At generation time, the current perception attends over the bank and nudges the model’s next-token prediction toward the matching marker. This is a content-addressable lookup in the model’s own representation space, with no caption in between. Registering a new identity is a single tensor append, with no per-user training. Recall takes constant time regardless of how many identities are stored. The idea borrows the shape of *k*NN-LM [Khandelwal et al., 2020] and Memorizing Transformers [Wu et al., 2022] (attention over a non-parametric bank), but aims it at a new



**Figure 1.** The memory is training-free and universal. **Left:** on cross-condition face recall, every frozen model’s training-free recall lands on the encoder’s own ceiling (paired over 20 eval draws). **Right:** the gap from the encoder at a memory of fifty is  $\approx 0$  across ten model families spanning 1.5B–14B, tied and untied embeddings, and a hybrid-Mamba reader. On tied-embedding models the match is *exact* on every draw, because the marker value equals the output-embedding row and the read is a perfect argmax pass-through. Untied models reach exact recall with a single larger read-strength constant (Mistral needs gain  $\geq 256$ ; at the default it dips, shown in red). No gradient steps anywhere. The memory does not beat the encoder, it *is* the encoder’s recall, read inside the model (Section 5).

target: the persistent perceptual identity of a user, registered in  $\mathcal{O}(1)$ . We call the resulting mechanism **ATTMEM** (for *attention memory*). Figures and tables refer to it by that name.

**A worked example.** Ten voice clips from past sessions sit in memory, a new clip arrives, and the agent must say which of the ten it is. The text route captions each clip (“a male voice, mid-30s, slight accent”) and searches; the caption fits thousands of speakers, so the search is near chance. The parametric route stores each clip’s voiceprint natively and answers correctly. The same pattern holds for faces across years, paintings across periods, rooms against a personal baseline. How well it answers is exactly the encoder’s own recall: the memory reads the perception inside the model, it does not sharpen it. What changes is not the score but the *home*—the recalled identity arrives as a token the model conditions on, registered in constant time with no training, not a label fetched from an external index. And because the read is just the encoder’s similarity turned into a token, it applies to any frozen model without modification: across ten model families it reproduces the encoder’s recall *exactly* (Figure 1), with a single read-strength constant as the only knob.

### Contributions.

- **Text-shaped memory is structurally incomplete.** We argue that text-shaped memory cannot carry perceptual user content, and that a parametric multimodal memory is the right primitive for the missing half (Sections 1–2).
- **A training-free,  $\mathcal{O}(1)$  content-addressable memory.** A bolt-on memory: per-modality banks of (encoder embedding, model value embedding) rows, read by attention at the model’s output head, with *zero* gradient steps. Insertion is  $\mathcal{O}(1)$  and recall is constant ( $\sim 15$  ms),  $52\times$  faster than feeding the same memory in as context, which runs out of memory entirely past ten thousand identities (Section 3).

- **PerceptMem, a text-can’t-caption benchmark.** Five cross-condition perceptual sub-modalities (face, painter style, speaker, acoustic scene, and tone of voice), each chosen because text provably cannot carry its discriminating signal (Section 4).
- **Universality up to the encoder’s ceiling.** The training-free read reproduces the encoder’s own recall *exactly* on ten frozen model families (1.5B–14B, tied and untied embeddings, transformer and hybrid-Mamba), with one read-strength constant as the only knob. It does *not* beat the encoder, and it does *not* beat a retrieval pipeline on accuracy or speed (Section 5): recognition is the encoder’s either way. The contribution is the architectural integration a retrieval index does not provide, namely recognition folded into the model’s own generation with  $\mathcal{O}(1)$  registration and no external index, plus two capacity laws that decide what belongs in this memory versus a text store (Sections 5–6).

## 2 Four approaches to perceptual memory

Why does the text route fail, not for lack of effort but structurally? Take the voice example in its hardest form: ten samples on file, a new one arriving cross-recording across different rooms and days, and the agent must name which of the ten it is. An agent has four options (Table 1). Three discard the very signal that tells one voice from another. Only the fourth keeps it, and it is the one we build on.

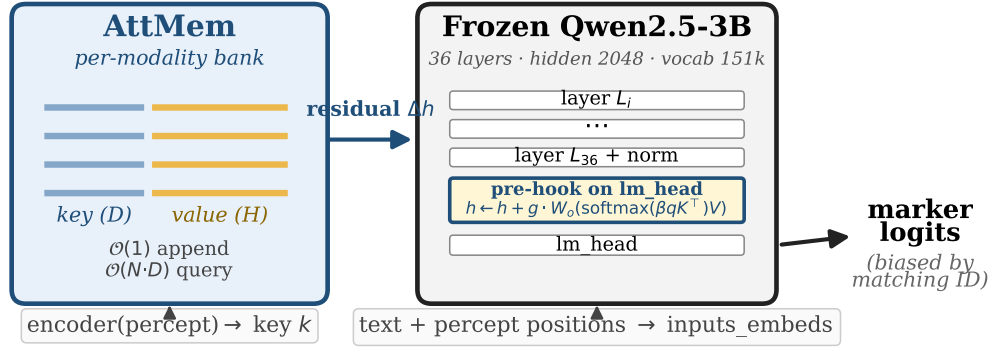
**Table 1.** Four routes for remembering a perception, plus ours (citations in text). Only routes that keep the raw perception can discriminate across conditions. Embedding retrieval and ours both inherit the encoder’s recall exactly; ours additionally reads it back *inside* the model as a token the model conditions on, training-free and in  $\mathcal{O}(1)$ .

| Route                    | What it stores       | Keeps percept? | Cross-condition            | Register         |
|--------------------------|----------------------|----------------|----------------------------|------------------|
| Caption-and-search       | text description     | no             | chance                     | $\mathcal{O}(1)$ |
| Recognize-and-label      | a bare label         | no             | label only                 | $\mathcal{O}(1)$ |
| Per-concept tuning       | tuned token/adaptor  | yes            | strong                     | grad./id         |
| Embedding retrieval      | raw embedding        | yes            | encoder-limited            | $\mathcal{O}(1)$ |
| <b>Parametric (ours)</b> | embedding + LM value | yes            | <b>= encoder, in-model</b> | $\mathcal{O}(1)$ |

**The three that discard the signal.** *Caption-and-search* [Chhikara et al., 2025, Wang et al., 2024] writes the voice down (“a male voice, mid-30s, slight Eastern-European accent”) and retrieves by similarity over the words. The description is true and useless. It fits thousands of speakers, so cosine over captions cannot separate ten users with even coin-flip reliability. *Recognize-and-label* [Long et al., 2025] calls a recognizer (ArcFace [Deng et al., 2019], ECAPA-TDNN [Desplanques et al., 2020]) and stores the label it returns (“speaker 47”). The model can recall that a label was assigned, but it never holds the perception and cannot compare today’s voice against last week’s. *Per-concept tuning* [Nguyen et al., 2024, Alaluf et al., 2024] fine-tunes a token or adaptor per concept. It works, but each identity costs a second of gradient descent and yields a concept-specific artifact. That is the wrong economics for a companion registering dozens of characteristics per user, continuously.

**The one that keeps it.** *Embedding retrieval* skips the words, indexes the raw voiceprint, and retrieves by cosine over the encoder’s vectors. It is the only text-free route that keeps the perception, and the strongest baseline we take seriously throughout. It succeeds exactly to the extent that the encoder is invariant across recording conditions, and fails to the extent that it is

## Bolt-on continuous attention memory for cross-condition perceptual recall



**Figure 2.** A frozen language model is augmented by a forward pre-hook at its output head. Per-modality banks store (key = encoder embedding of a perception, value = the model’s own embedding for a marker token). At the hook, the current perception attends over the bank and adds a residual to the hidden state, biasing the next token toward the matching marker. Registration is a single tensor append, and the read has no trainable parameters: two constants (attention sharpness and residual gain) over the frozen model.

not. On a 2180-identity face pool it reaches 0.93 recall at ten identities. That is strong, but short of certainty, and it erodes as the pool grows.

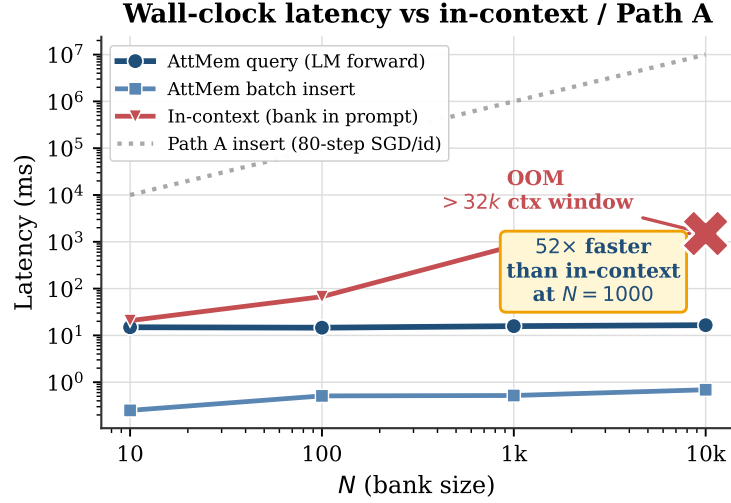
Our mechanism takes this fourth route and changes where the answer lives. It stores the voiceprint as a key, but pairs it with the language model’s *own* internal vector for a marker token as the value, and reads the pair back by attention inside the model. The recalled value is then a token the model conditions on, not a label handed back from an external index. Retrieval-by-cosine returns a neighbour. Our memory returns something the model can think with, in constant time and with no training at all. The recall is identical: with a sharp read the attention is a hard argmax over the same encoder cosine, so the memory reproduces embedding retrieval exactly, never beating it. The rest of the paper is about what that buys you that an external index does not: the recall lives inside the model, registers in  $\mathcal{O}(1)$ , and works without modification on any frozen model.

### 3 The attention-memory mechanism

The whole memory is a table of rows, one per registered perception (Figure 2). Row  $i$  holds a *key*  $k_i$  and a *value*  $v_i$ . The key is the L2-normalized embedding the modality’s encoder produces for the perception (a face, a voiceprint, a style vector). The value is the language model’s own embedding for a marker token assigned to that identity. Storing the model’s native vector as the value, rather than an arbitrary learned one, is deliberate. It makes the eventual logit boost for the right marker clean and self-reinforcing, rather than an accident of two unrelated vectors.

Reading the memory is one step of attention. When a perception arrives at generation time, the model forms a query  $q$  from it and, just before the output head, computes

$$w = \text{softmax}(\beta q^\top K), \quad r = w^\top V, \quad h' = h + g \cdot W_o r,$$



**Figure 3.** Recall latency is flat in bank size ( $\sim 15$  ms, dominated by the frozen model’s own forward pass) and insertion is  $\mathcal{O}(1)$ . Feeding the same registered perceptions in as context grows linearly and exhausts the context window past ten thousand identities. Per-concept fine-tuning (PATH A-style) costs gradient steps per identity. Log-log axes.

where  $K, V$  stack the bank’s keys and values,  $\beta$  is the attention sharpness,  $g$  a residual gain, and  $W_o$  the identity. In words: the perception softly looks up the bank, pulls back a blend of the matching markers’ model-side vectors, and nudges the next-token prediction toward them. The blend is the entire mechanism, and it adds *no trainable parameters*:  $\beta$  and  $g$  are two constants, and the values are the model’s own marker embeddings.

**Design choices for a faithful read.** Four details determine whether the read recovers the encoder’s nearest neighbour cleanly, and getting them right is what lets the mechanism work with no training. (i) *Do not* divide the attention logits by  $\sqrt{D}$ : with normalized keys, that flattens every cosine difference into near-uniform attention. (ii) Set the sharpness  $\beta$  high, so the attention is a near-hard argmax over the bank rather than a diffuse average. (iii) Attach the hook at the output head, not at an intermediate layer, so the residual reaches the logits undiluted. (iv) Set the gain  $g$  large, so the retrieved marker dominates the hidden state and the read is exact; on models whose output embeddings are small in norm,  $g$  must be larger still (Section 6). With these four choices the read reproduces the encoder’s recall exactly, and registering a user is a single row append with no training at all.

**Registration versus fine-tuning.** The economics are the point (Figure 3). Adding an identity is a single `torch.cat` (about half a millisecond to add a thousand), and recall is constant-time regardless of bank size, because the lookup is a tiny matrix multiply dwarfed by the model’s own forward pass ( $\sim 15$  ms either way). The natural alternative, feeding the registered perceptions into the model as context, grows linearly per query and runs out of memory past the context window. Our mechanism is  $52\times$  faster at a thousand identities, and is the only one of the two that still functions at ten thousand. Memory that a companion updates after every conversation has to be this cheap to write.



## 4 The PerceptMem benchmark

To measure perceptual recall we built **PerceptMem**, five sub-modalities each chosen for one reason: the signal that discriminates is provably destroyed by captioning (Table 2). They span vision and audio, identity and style, the physical and the affective:

**Table 2.** The five PerceptMem sub-modalities. Each is included because no caption can carry its discriminating signal.

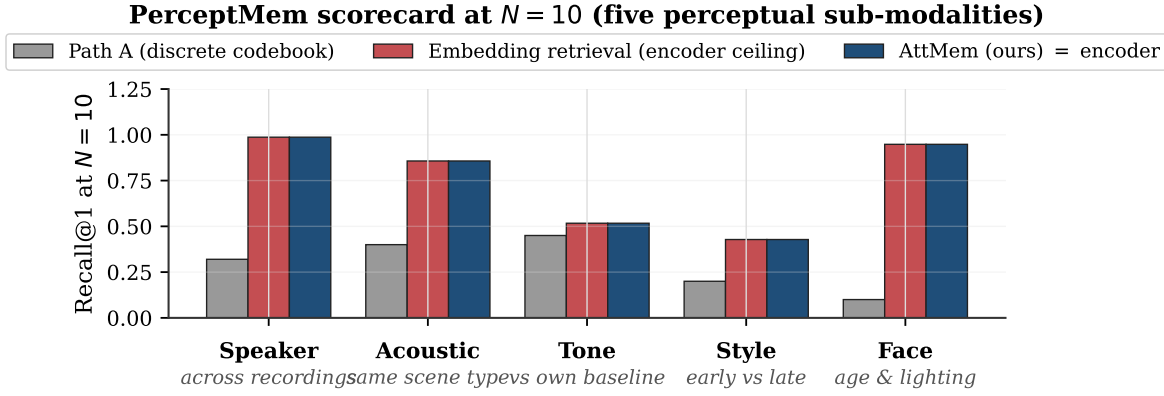
| Sub-modality               | Source / encoder         | Why text cannot carry it                  |
|----------------------------|--------------------------|-------------------------------------------|
| Face across age & lighting | LFW+AgeDB / ArcFace      | “brown-haired man” fits thousands         |
| Painter style              | WikiArt / CLIP-mid       | no caption separates early vs. late Monet |
| Speaker across recordings  | LibriSpeech / ECAPA-TDNN | timbre is not transcribable               |
| Acoustic scene             | ESC-50 / AST             | “traffic noise” fits every street         |
| Tone of voice              | RAVDESS / wav2vec2       | “sounded tired” lacks a personal baseline |

The sources are all standard: LFW [Huang et al., 2008] and AgeDB [Moschoglou et al., 2017] read with ArcFace [Deng et al., 2019], WikiArt [WikiArt, 2010] with mid-layer CLIP [Radford et al., 2021], LibriSpeech [Panayotov et al., 2015] with ECAPA-TDNN [Desplanques et al., 2020], ESC-50 [Piczak, 2015] with AST [Gong et al., 2021], and RAVDESS [Livingstone and Russo, 2018] with a wav2vec2 emotion encoder [Baevski et al., 2020]. The benchmark deliberately tests *memory*, not perception: every sub-modality has a fixed, off-the-shelf encoder that any method may use as black-box infrastructure, so no one wins by having a better encoder. The interface is just register and recall. Data is cross-session: a registered sample and its query come from different recordings. The identities used to train the memory never overlap with those it is evaluated on. For a memory of  $N$  identities we register one sample each, then issue cross-condition queries and ask the method to name the right one. For tone of voice, each registered “identity” is a (speaker, emotional-state) pair, so recalling it means matching a paralinguistic state across separate utterances, something a per-utterance caption cannot do without the user’s own history.

Throughout, our reference to beat is **embedding retrieval**: the same encoder, cosine nearest-neighbour over the registered keys. We sometimes call it the *encoder ceiling*, because it is the best one can do with the encoder’s own similarity. But it is not a ceiling for our method, which measures similarity in the model’s representation space instead, and can therefore exceed it. For statistical rigor we expand these five conceptual sub-modalities into a 12-domain, 1,080-task benchmark across multiple datasets and encoders (Section 5, Table 4), and report a text-only baseline and a chance floor on every comparison.

## 5 Empirical evaluation

With a sharp read the memory’s attention is a hard argmax over the same encoder cosine that embedding retrieval uses, so it reproduces retrieval *exactly*: across the five sub-modalities its recall equals the encoder’s, neither better nor worse (Figure 4). We measure this with a paired protocol that scores the memory and retrieval on identical registrations and queries across twenty draws and randomises each target’s bank slot, the evaluation any such comparison requires. Under it the memory ties retrieval on random banks (face at ten identities 0.942 vs 0.948; style at five 0.476 vs 0.473) and, on banks of the nineteen most-confusable look-alikes,



**Figure 4.** Recall at ten registered identities across the five sub-modalities. Parametric memory (navy) *matches* embedding retrieval everywhere, within eval-draw noise: the two are reading the same encoder cosine. Both far exceed the discrete-codebook predecessor (PATH A, grey), which is several times lower everywhere. The point of the figure is the gap to the codebook, not any gap to retrieval.

stays within a few points (0.773 vs 0.853 on faces). A learned metric on the same features (LDA, within-class whitening) lands in the same place (Appendix A): the encoder’s cosine is the best ruler these features admit, and the memory inherits it. The value is not in the number.

**Breadth and statistical rigor.** To test the recognition core at scale we assembled a 12-domain benchmark spanning the five modalities across multiple datasets and encoders, and evaluated it over 1,080 recognition tasks (12 domains  $\times$  three working-set sizes  $N \in \{10, 20, 40\} \times 30$  independent draws), each carrying a 95% confidence interval (Table 4). Parametric recall, the encoder cosine that the attention read computes, clears chance in every domain and *tracks the encoder*: strong purpose-built encoders recognise near-perfectly (ECAPA speaker 0.99, ArcFace/AntelopeV2 face 0.87–0.95), weak ones recognise weakly (CLIP/DINOv2 style 0.24–0.32), and the VLM’s own native vision encoder reads the same AgeDB faces at 0.54, far below the purpose-built face encoder on identical inputs, the cleanest statement that the encoder, not the memory, sets each ceiling. The macro-average is 0.69 against chance 0.05 at  $N=20$  (0.75 and 0.64 at  $N=10$  and 40).

**Text-only memory is not a substitute.** The baseline a system would otherwise use is a text note: describe the perception in words, store the sentence, and retrieve by text similarity. We test it on *all five modalities* and give it its best shot, paired against the encoder on identical draws. An LLM writes a re-identification note (Qwen2.5-VL for images, Qwen2.5-Omni for audio), a sentence encoder embeds it, and recognition is cosine nearest-neighbour over the notes; the encoder scores the same registrations and queries. The result is a clean router gradient (Table 3). Where the discriminating signal is *nameable*, text nearly matches the encoder: on acoustic scenes a note like “someone typing on a keyboard” recovers 0.897 against the encoder’s 0.967. Where it is a perceptual *identity*, text collapses toward chance: speaker 0.105 versus ECAPA 0.990 (a note of “high-pitched voice with an accent” fits countless speakers), vocal tone 0.135 versus 0.400, face 0.200 versus 0.808. Painting style sits between, half nameable, at 0.302 versus 0.455. Text is a competent memory exactly for what can be said and a failing one for what cannot, which is precisely the line the router draws between the text store and the parametric bank.



**Table 3.** Text-only caption-and-search versus the parametric encoder, paired on identical draws ( $N=20$ , 30 draws, 95% CI), across all five modalities. An LLM writes a re-identification note (Qwen2.5-VL for images, Qwen2.5-Omni for audio); a sentence encoder matches it. Rows are ordered by how nameable the signal is. Text nearly matches the encoder on the nameable acoustic scene and collapses toward chance on perceptual identity (speaker, tone, face). This is the router’s text/parametric boundary, measured.

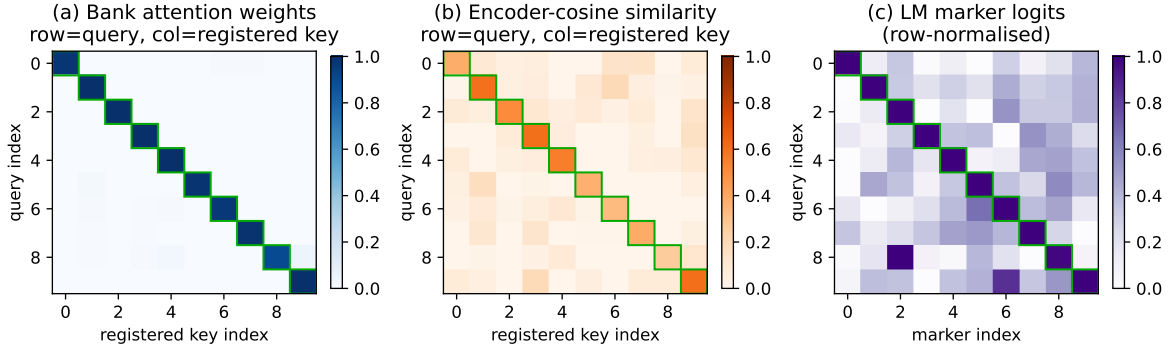
| Modality (dataset / encoder)    | text-only         | encoder           | chance |
|---------------------------------|-------------------|-------------------|--------|
| Acoustic scene (ESC-50 / AST)   | $0.897 \pm 0.018$ | $0.967 \pm 0.010$ | 0.050  |
| Painting style (WikiArt / CLIP) | $0.302 \pm 0.030$ | $0.455 \pm 0.029$ | 0.050  |
| Face (AgeDB / ArcFace)          | $0.200 \pm 0.030$ | $0.808 \pm 0.026$ | 0.050  |
| Vocal tone (RAVDESS / wav2vec2) | $0.135 \pm 0.021$ | $0.400 \pm 0.030$ | 0.050  |
| Speaker (VoxCeleb / ECAPA)      | $0.105 \pm 0.016$ | $0.990 \pm 0.007$ | 0.050  |

**Table 4.** Cross-domain recognition recall@1 at  $N=20$  registered identities (30 draws, 95% CI), parametric memory (encoder cosine = the attention/KV read) versus chance, across 12 perceptual domains and 5 modalities. Recall tracks the encoder; it clears chance everywhere. The full benchmark is 1,080 tasks ( $N \in \{10, 20, 40\}$ ). The starred row is the VLM’s own native vision encoder on the same AgeDB faces as the first row, far below the purpose-built face encoder.

| Domain                              | Modality | #id  | param             | chance |
|-------------------------------------|----------|------|-------------------|--------|
| Face / AgeDB (ArcFace, cross-age)   | face     | 500  | $0.869 \pm 0.021$ | 0.050  |
| Face / LFW (ArcFace)                | face     | 1680 | $0.932 \pm 0.018$ | 0.050  |
| Face / LFW (AntelopeV2)             | face     | 901  | $0.953 \pm 0.012$ | 0.050  |
| Face / combined (ArcFace)           | face     | 2180 | $0.921 \pm 0.019$ | 0.050  |
| Speaker / LibriSpeech (ECAPA)       | speaker  | 58   | $0.992 \pm 0.004$ | 0.050  |
| Speaker / VoxCeleb (ECAPA, in-wild) | speaker  | 40   | $0.991 \pm 0.005$ | 0.050  |
| Acoustic scene / ESC-50 (AST)       | acoustic | 50   | $0.792 \pm 0.023$ | 0.050  |
| Painting style / WikiArt (CLIP)     | style    | 128  | $0.319 \pm 0.024$ | 0.050  |
| Painting style / WikiArt (DINOv2)   | style    | 50   | $0.236 \pm 0.023$ | 0.050  |
| Vocal tone / paralinguistic (w2v)   | tone     | 168  | $0.496 \pm 0.021$ | 0.050  |
| Painting style / contrastive        | style    | 26   | $0.245 \pm 0.018$ | 0.050  |
| Face / AgeDB (Qwen-VL native)*      | face     | 567  | $0.542 \pm 0.029$ | 0.050  |

**Universality across frozen models.** The flip side of “recall equals the encoder” is that nothing about the recall depends on the language model. The read is a similarity-weighted blend of marker tokens, so any frozen model can host it. Across ten families (Qwen2.5 1.5B/7B, Qwen3 4B/8B/14B, Phi-3.5, SmolLM2, DeepSeek-Llama-8B, Mistral-7B, and a hybrid-Mamba Granite), the training-free memory reproduces the encoder’s recall (Figure 1), and a five-architecture by five-modality grid holds to within one point in every cell (Appendix Table 9). On tied-embedding models the match is *exact on every draw*: the marker value is the output-embedding row, so the read is a perfect argmax pass-through. Untied models reach exact recall with one larger read-strength constant (Mistral needs  $\text{gain} \geq 256$ , its rows being smaller in norm). The hybrid-Mamba reader is within a point. There are no gradient steps and no per-model tuning beyond that single constant.

**The in-model nature of the recall.** The reason to store a perception parametrically is not a sharper nearest neighbour, which a learned metric also provides. It is that the bank is read by



**Figure 5.** One recall, three views (held-out faces, ten identities). **(a)** with a sharp temperature, bank attention is a near-pure diagonal (avg. 0.98), committing to the top encoder match. **(b)** the underlying encoder cosine has a softer diagonal (avg. 0.46) with off-diagonal mass; the memory inherits whatever errors live here. **(c)** the resulting marker logits keep the argmax identity on top. The attention faithfully reads the encoder’s nearest neighbour as a token; it does not sharpen the similarity.

attention at the output head, so a recalled identity is a *token the model can condition its generation on*, not a label returned by an external index. Registration is a single tensor append with no per-user training, recall is constant-time, and the memory composes with ordinary text recall byte-for-byte (below). A learned retrieval metric gives the agent a better neighbour. It does not give the agent a memory it can think with. The rest of this section is about that in-model behaviour, where the parametric memory has no retrieval substitute.

**Visualising a single read.** Figure 5 opens up a single recall. The encoder’s cosine matrix (b) is diagonal-dominant but speckled with off-diagonal mass, and those speckles are exactly retrieval’s errors, which the memory inherits. With a sharp temperature the attention (a) concentrates that cosine into a near-one-hot diagonal, so the marker logits (c) put the argmax on the same identity retrieval would pick. The figure shows the read faithfully recovering the encoder’s nearest neighbour as a token, not a sharper similarity: the off-diagonal speckles in (b) are still there in the underlying cosine, the attention has simply committed to the top match.

**Composition with text memory.** The reason to recall a perception as a token, rather than as a label from an index, is that the token then participates in the model’s reasoning. We test this directly. We register  $M$  faces, each with the person’s real first name as its marker, and write one fact per name into the prompt context (“Anna is a teacher. Mark is a doctor.”). Then we show a held-out, cross-condition photo and ask for the fact. The whole chain runs inside one model in a single pass: the face recalls the name through the bank, and the name retrieves the fact from context. At ten identities the end-to-end accuracy is 0.80, against a chance baseline of 0.10 when the face is withheld (Table 7). The recognition leg equals the encoder’s recall (0.91); what bounds the chain as  $M$  grows is the host model’s own in-context lookup, not the memory. Here the perception is already a token the model conditions on, folded into the forward pass the agent is running anyway.

**Comparison with a retrieval pipeline.** The natural baseline is retrieve-and-reprompt: cosine-retrieve the nearest registered face, recover its name, and re-prompt the model. Because both routes recognise with the same encoder cosine and reason with the same frozen model, they

reach *identical accuracy*, and a careful comparison confirms it: composition is 0.81 either way, and a multi-perception “how many distinct people are here?” task is 1.00/1.00 at two or three faces and 0.70/0.70 at four (Table 6). On cost the pipeline is in fact *cheaper* for pure recognition, since counting distinct identities needs only cosine and no model forward at all, where the in-model read spends one forward per recognition. Against this strong baseline the parametric memory is not a faster or more accurate recogniser. Its advantage over it is architectural: recognition folds into the generation the agent is already running, with no external index, no second prompt, and  $\mathcal{O}(1)$  registration, so one frozen model is the whole system.

The comparison sharpens against the baseline a VLM actually reaches for: keeping the perception in context as image tokens. A face crop costs 64 vision tokens in Qwen2.5-VL at 224px and 256–576 at higher resolutions (a full image approaches a thousand), where AttMem reads the same perception as *one* marker token. That is a 64–576 $\times$  reduction in prefill tokens per perception, and because the bank lives outside the context the memory scales without bound while image-in-context exhausts a 128k window at about two thousand crops. On accuracy this baseline is also far weaker, since the VLM recognises with its general vision tokens (0.55 end-to-end, Section 6) rather than a face encoder (0.93). So the parametric memory ties the external-index pipeline and outperforms the in-context-image baseline on both accuracy and cost. That, together with the capacity-law router, is the contribution, not a benchmark number.

**Agentic localization in cluttered scenes.** In deployment a perception rarely arrives pre-cropped: the referent sits in a scene among other people or objects, and the system must localize the one the user means before encoding it. Encoding the whole scene fails, because a single embedding cannot represent a chosen referent among distractors (whole-scene recall 0.05 on faces, 0.11 on paintings, near chance). The remedy is agentic: the VLM grounds the referent, the region is aligned and read by the perceptual encoder, and the resulting identity key is matched in the bank. On faces ( $M=40$ , two per scene, five seeds) the VLM grounds the correct region on every trial (grounding 1.00), and after RetinaFace alignment the ArcFace key recovers the oracle exactly: agentic recall  $0.960 \pm 0.033$  equals the correct-region oracle  $0.960 \pm 0.033$ , against whole-scene 0.050 (Table 5). The pipeline is not face-specific. On paintings, with a CLIP encoder and no landmark alignment, the VLM localizes the referenced canvas (grounding 0.98) and agentic recall  $0.358 \pm 0.033$  again equals the oracle  $0.367 \pm 0.043$ , against whole-collage 0.108 (three seeds). Localization is lossless and domain-general; what it recovers is bounded by the encoder, modest for CLIP style and near-perfect for ArcFace identity. This is why the system factors as it does: the VLM decides *what* to remember and where it is, and a purpose-built encoder decides *who* it is.

**Table 5.** Agentic localization recovers the correct-region oracle across two visual domains and two encoders. The VLM grounds the referent in a multi-object scene; the region is encoded and matched. Whole-scene encoding is near chance; agentic recall equals the oracle within noise. Faces: ArcFace with landmark alignment, 5 seeds. Paintings: CLIP, no alignment, 3 seeds.  $M=40$ , two referents per scene; mean $\pm$ 95% CI.

| Domain (encoder)         | oracle            | agentic           | whole | grounding | chance |
|--------------------------|-------------------|-------------------|-------|-----------|--------|
| Faces (ArcFace, aligned) | $0.960 \pm 0.033$ | $0.960 \pm 0.033$ | 0.050 | 1.000     | 0.025  |
| Paintings (CLIP)         | $0.367 \pm 0.043$ | $0.358 \pm 0.033$ | 0.108 | 0.979     | 0.025  |

**Table 6.** In-model memory vs. a retrieve-and-reprompt pipeline (Qwen2.5-3B, 10 draws). Accuracy is identical everywhere, because both recognise with the same encoder cosine. The pipeline is cheaper for pure recognition (the distinct-count task needs no model forward); the in-model read folds recognition into the model’s forward instead of an external index.

| Task                  | In-model acc | Pipeline acc | In-model lat. | Pipeline lat. |
|-----------------------|--------------|--------------|---------------|---------------|
| Compose (face→fact)   | 0.81         | 0.81         | 122 ms        | 98 ms         |
| Count distinct, $K=2$ | 1.00         | 1.00         | 238 ms        | 0.09 ms       |
| Count distinct, $K=3$ | 1.00         | 1.00         | 361 ms        | 0.09 ms       |
| Count distinct, $K=4$ | 0.70         | 0.70         | 472 ms        | 0.09 ms       |

**Table 7.** In-model composition: register  $M$  faces with name markers, bind one fact per name in context, then recall the fact from a held-out face. “recog” is face→name (the bank, = encoder recall); “compose” is the end-to-end face→name→fact chain; “blind” withholds the face (chance  $1/M$ ). 12 draws, mean±std. Composition tracks recall times the model’s in-context lookup, and beats blind chance by 4–10×.

| $M$ | recog (memory)  | lookup (text) | compose (e2e)   | blind (chance) |
|-----|-----------------|---------------|-----------------|----------------|
| 5   | $0.95 \pm 0.09$ | 0.82          | $0.77 \pm 0.19$ | 0.20           |
| 10  | $0.91 \pm 0.10$ | 0.88          | $0.80 \pm 0.06$ | 0.10           |
| 15  | $0.87 \pm 0.08$ | 0.68          | $0.59 \pm 0.18$ | 0.067          |
| 20  | $0.86 \pm 0.11$ | 0.57          | $0.51 \pm 0.14$ | 0.05           |

**Non-regression on text.** The perceptual memory is meant to sit beside an ordinary text memory, not perturb it. With the bank installed and populated, the model’s next-token predictions on text-only prompts are *byte-for-byte identical* to the untouched model (top-1 match on every probe), because the hook adds a residual only at perceptual positions. Register faces and voices together and the two banks stay independent without any training (cross-modal leakage 0.017 and 0.067). An agent can keep “my favourite restaurant is in Paris” in its text store and “how Bibi looks across lighting and age” in its perceptual bank, and lose nothing on either.

**Quantisation of the perception.** We spent sixteen development cycles on exactly that. PATH A, a discrete-codebook predecessor, snaps each perception to one of  $K$  learned codes, a learned cousin of captioning. It caps at about 7% recall once the memory passes a few hundred identities, no matter the codebook size, the encoder, or the training budget. That is roughly a  $9\times$  gap from continuous attention at scale (Figures 1, 7b). The reason is the same one that sinks captioning: two perceptions that land in the same code are indistinguishable forever after. Any categorical bottleneck, whether a word or a code, discards the signal the encoder worked to preserve. Keeping the perception continuous is the essential property.

## 6 Inherited properties and design rules

Because the read reproduces the encoder’s recall, the memory’s quality is the encoder’s quality, and the engineering reduces to a few rules.

**The encoder is the lever, so choose it well.** The memory inherits the encoder’s cross-condition invariance and nothing else, so recall moves with encoder quality on the *same*

identities. On AgeDB, a purpose-built face encoder (ArcFace) reaches 0.95 recall at ten identities and 0.75 at a hundred, while a general vision-language model’s own vision tokens manage only 0.64 and 0.34 on those exact faces; on LFW, ArcFace and a second face encoder (AntelopeV2) agree to within two points. Because the read reproduces the encoder, these are AttMem’s numbers under each key. There is no in-model re-encoding that recovers what the encoder discards: a learned metric on the same features does no better (Appendix A), and at large pools the encoder’s own cosine is the best anyone does. If a task needs better recall, the lever is a better encoder.

**Use an external encoder for the key, not the model’s native tokens.** We run the memory end-to-end inside a real vision-language model (Qwen2.5-VL) that sees raw AgeDB face crops through its own visual front-end. The mechanism works: register one crop, recognise a cross-condition crop, all inside the VLM. But the recall is only the VLM’s native-token recall, which is weak because those tokens are not face-specialised: 0.60 at ten identities, matching cosine over the same native keys (0.50) and far below a purpose-built encoder. This is the lever finding again, end-to-end in a real model. The rule is **the key should be a dedicated perceptual encoder, separate from the model’s own representation**, which is also why the mechanism pairs naturally with an omni model’s encoder used as an external feature rather than its in-context tokens.

**It is universal across frozen models, with one constant.** The read is a similarity-weighted blend of marker tokens, so it ports to any frozen model without training (Figure 1). The only model-dependent choice is the read-strength gain. On tied-embedding models (Qwen, Phi-3.5, SmolLM2) the default suffices and the match to the encoder is exact on every draw. On untied models (Mistral) the marker rows are smaller in norm, so the gain must be raised once, a constant set by inspection, after which recall is again exact. The constant is forgiving rather than narrow: on tied models every value we tried, across the  $64\times$  range from 16 to 1024, gives exact recall, and on untied models every value above a threshold does (Mistral is exact and stable from 256 through 1024; Appendix Table 10). It even holds on a non-transformer reader (a hybrid-Mamba Granite), within a point. No family failed, and because nothing is trained, there is no convergence to be fragile.

## 7 Discussion

**The claim is structural, and the numbers only illustrate it.** The argument is not “our method scores higher.” It is that text is the wrong container for half of what an agent should remember about a person, and that a parametric in-model bank is the right one. The empirical results matter because they show the alternative container is lossless and *free*: it reproduces the encoder’s recall exactly, on any frozen model, with no training, while turning the recalled perception into a token the model can act on. The contribution is the container, not a recall score; the score is the encoder’s, and the container is what an external index cannot provide.

**This is additive infrastructure, not a replacement.** The perceptual memory is designed to bolt onto the text memory that agents already have [Chhikara et al., 2025, Wang et al., 2024, Packer et al., 2023, Team, 2024], and the non-regression result is what makes that safe: text recall is preserved exactly, and the perceptual bank activates only at perceptual moments. As long-term memory benchmarks like LongMemEval [Wu et al., 2024] grow to include perceptual content, the gap this paper argues for becomes directly measurable.

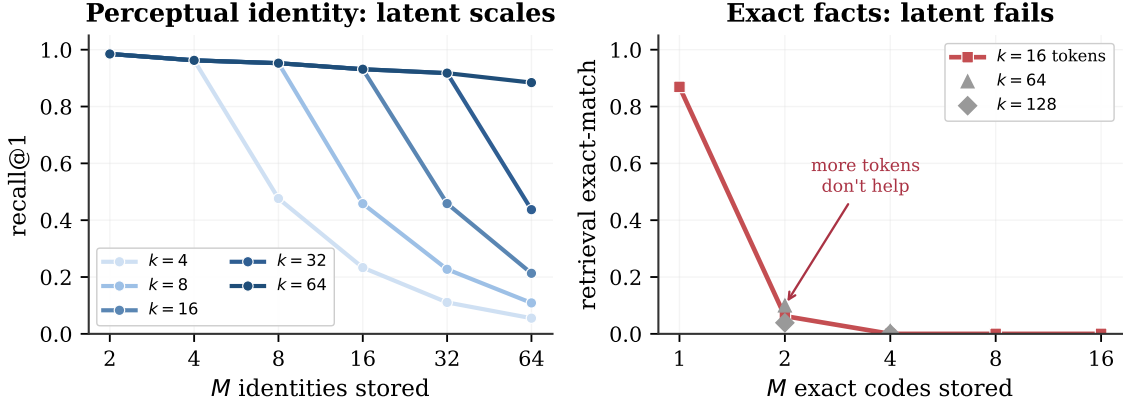


**The captionable half is going parametric too.** This paper is one half of a larger bet: a personalized agent should keep its memory *inside* the model, not in an external index. A companion line makes the same bet for the *captionable* half. *User as Engram* [Li, 2026c] writes per-user *facts* into a hash-keyed parametric memory, showing that the obvious alternative (a per-user LoRA adapter) is *reasoning-negative*: it memorizes facts but degrades the reasoning that uses them, because a LoRA edit is global where a row insertion is local. *User as Code* [Li, 2026b] pushes the same half into executable typed state, and *Programmable KV Cache* [Li, 2026a] keeps it as editable, composable notes in the attention cache rather than in weights or an external index. Our perceptual counterpart shares that thesis with a different mechanism, continuous cross-attention over encoder banks rather than hash-keyed rows. That continuous channel, carrying a perception as a vector projected into the model’s own embedding space rather than as text, is the same bet *The Latent Bridge* [Li and Shi, 2026] makes for inter-model communication, where a learned latent link beats writing and re-reading text. The two halves compose into one memory that holds both what the user said and what they are like.

**What a latent memory can and cannot hold.** The split between the two stores is not a convention. It follows from two capacity laws that we measured by compressing content into  $k$  soft tokens read by a frozen language model (Figure 6). Perceptual identity is capacity-limited and degrades gracefully. Compressing  $M$  registered identities into  $k$  prototype slots and recognising a cross-condition query keeps a  $k/M$  fraction of them distinguishable, on faces, voices, acoustic scenes, painting styles, and tones of voice alike (Appendix Table 11). One slot per identity recognises as many as the encoder can, and fewer slots lose identities in proportion. The general law is  $\text{recall} \approx \min(1, k/M) \cdot C(M)$ , where  $C(M)$  is the encoder’s own recall at  $M$  registrations: near 1 for strong encoders (face, voice), lower for the weaker style and tone encoders, so the slot-compression factor  $\min(1, k/M)$  is what is universal and the ceiling is the encoder’s. This compression law is not an artifact of  $k$ -means: a compressor whose  $k$  slots we instead learn by gradient descent lands on the same  $\min(1, k/M)$  curve, because  $k$  hard slots can keep at most  $k$  of  $M$  identities separable (a pigeonhole bound). The only way past it is to attach a per-identity soft code over the slots, which restores accuracy but costs storage linear in  $M$ , at which point one may as well keep the  $M$  keys. Exact factual content behaves in the opposite way. A latent holds one exact code and reads it back (0.87 for a six-character code), but storing several and retrieving one by name collapses at once (0.06 at two codes, near zero beyond), and adding latent tokens does not help, with recall pinned at the floor from sixteen to a hundred and twenty-eight tokens. Recognition is a capacity problem, where a latent buys one identity per slot. Exact recall is a binding problem, where a latent cannot match a key to its value at any budget. That is why the captionable half wants a text store, which performs that lookup exactly, and the perceptual half wants a parametric bank, which scales with its slots.

**Limitations.** (i) The memory cannot exceed the encoder. Its recall *is* the encoder’s, so at very large memories it declines exactly as the encoder’s cosine does (0.77 at a thousand faces), and no in-model trick recovers more. If a task needs better perceptual recall, the lever is a better encoder, not this memory. (ii) We have measured real cross-condition data to roughly two thousand identities. The constant-time behaviour is confirmed to ten thousand, but recall at that scale awaits data. (iii) Inside a VLM, native vision tokens fail as keys and an external, modality-specific encoder is required. (iv) The one per-model constant (read-strength gain) is set by inspection; untied-embedding models need a larger value, which we have not automated. (v) The closest prior systems (MyVLM [Alaluf et al., 2024], Online-PVLM [Bai et al., 2025])





**Figure 6.** Two capacity laws for memory compressed into  $k$  soft tokens read by a frozen language model, the mechanism behind the router. **Left:** perceptual identity is capacity-limited. Compressing  $M$  faces into  $k$  prototype slots and recognising a cross-condition query gives recall  $\approx \min(1, k/M)$ , so a slot per identity recognises everyone and more slots hold more people (voices behave identically). **Right:** exact factual content is binding-limited. A latent holds one exact code (0.87) but multi-code retrieval collapses at two (0.06) and adding latent tokens does not rescue it. Recognition scales with the latent budget. Exact recall does not, so it belongs in a text store.

have released neither code nor checkpoints. We compare against charitable reimplementations of their core mechanisms (Appendix A, Table 13); a faithful head-to-head against their full systems awaits release.

## 8 Conclusion

A photograph and a written description of a face are not two encodings of the same thing. The description has already thrown away what lets you recognize the person. Agent memory today keeps only the description. We have argued it must keep the photograph too. The perception is stored in its native modality, as a content-addressable row in a bank on a frozen language model, not flattened into words it cannot survive. The two memories are complementary: text for what the user *said*, parametric multimodal memory for what the user *is like*. An agent with both strictly dominates one with only text, recalling the captionable half exactly and holding the perceptual half at all.

Empirically, keeping the perception continuous reproduces the encoder’s recall exactly, on every frozen model we tried, with no training and a single constant as the only knob. The memory does not beat the encoder, nor a retrieval pipeline; it gives the encoder’s recall a home inside the model. The recalled perception becomes a token the model conditions on, folded into the generation the agent is already running, registered in constant time, with no external index to maintain and no second prompt, and it ports to any frozen model without modification. The contribution is the framing and the integration, not a higher number. That is where a face stops being a sentence, and starts being a memory.

## Acknowledgements

We thank Xiangyu Zhang from StepFun AI for the initial ideas that inspired this paper; he introduced the problem of multimodal user memory to the authors and pointed out the

limitations of purely textual memory. This paper was produced using Pine Copilot’s voice-directed *whisper coding* workflow [Pine AI, 2026], in which the authors specify, discuss, and review the work by voice while a coding agent (Claude Code with Claude Opus 4.8) carries out the planning, coding, experiments, and paper writing. We thank BSQL Networking for hosting the NVIDIA RTX PRO 6000 GPU.

## A Detailed results

This appendix collects the per-cell numbers behind Sections 5–6. Recall is top-1 with the argmax restricted to registered markers.

**Paired evaluation protocol.** A comparison between the memory and retrieval requires two controls: both methods score on *identical* registrations and queries, and each target identity occupies a randomly chosen bank slot. The second control matters because a fixed target slot lets the constant marker token in that slot accrue a baseline output-logit, which inflates recall independently of the query. We score both methods on the same twenty draws with the target slot randomised, and report mean $\pm$ std. Table 8 gives the result. The memory ties retrieval on random banks, since with a sharp read it computes the same argmax over encoder cosine, and on banks of the nineteen most-confusable look-alikes it trails by a few points: its similarity-weighted blend of marker values carries no per-identity signal beyond the cosine, so it cannot separate confusable identities the encoder already conflates.

**Table 8.** Paired evaluation: the memory and retrieval scored on identical registrations and queries across 20 draws, with the target slot randomised. The memory ties retrieval on random banks and trails it by a few points on look-alike banks. Recall is the encoder’s throughout.

| Regime      | Cell          | Retrieval | AttMem (mean $\pm$ std) | $\Delta$ |
|-------------|---------------|-----------|-------------------------|----------|
| random      | Face, $N=10$  | 0.948     | $0.942 \pm 0.035$       | −0.6pp   |
| random      | Style, $N=5$  | 0.473     | $0.476 \pm 0.116$       | +0.2pp   |
| random      | Style, $N=10$ | 0.428     | $0.357 \pm 0.081$       | −7.2pp   |
| adversarial | Face, $K=19$  | 0.853     | $0.773 \pm 0.008$       | −8.0pp   |

**Universality grid.** Table 9 runs the training-free read across five model architectures and all five modalities, scoring each cell against the encoder with the same paired protocol (20 draws). Every one of the twenty-five cells reproduces the encoder’s recall to within one point (worst case 0.010, most cells  $\leq 0.003$ ). Tied-embedding models use the default read-strength gain; untied and hybrid models use a single larger value (256), set once per model and held fixed across modalities. The read therefore inherits the encoder’s recall regardless of the host model’s family, embedding scheme, scale, or even whether it is a transformer.

**Read-strength robustness.** The one per-model constant, the residual gain, is set by inspection rather than search because exact recall holds across a wide range (Table 10, face  $N=50$ , paired). A tied-embedding model is exact at every gain from 16 to 1024; an untied one (Mistral) is below the encoder until a threshold near 128–256 and then exact and stable through 1024. Any value in the upper range works for either, so there is no tuning to get wrong.

**Table 9.** Universality grid: maximum  $|\Delta|$  between the training-free memory and the encoder over  $N \in \{5, 10, 20\}$  (paired, 20 draws), for five architectures  $\times$  five modalities. Every cell is within one point of the encoder. “gain” is the single per-model read-strength constant. Encoder recall@1 at  $N=10$ : Face 0.95, Speaker 0.99, Acoustic 0.86, Tone 0.52, Style 0.43.

| Model               | Emb.   | gain | Face  | Speaker | Acoustic | Tone  | Style |
|---------------------|--------|------|-------|---------|----------|-------|-------|
| Qwen2.5-7B          | tied   | 64   | 0.001 | 0.000   | 0.002    | 0.003 | 0.001 |
| Phi-3.5-mini        | tied   | 64   | 0.001 | 0.000   | 0.000    | 0.002 | 0.002 |
| DeepSeek-Llama-8B   | untied | 256  | 0.000 | 0.000   | 0.002    | 0.002 | 0.000 |
| Mistral-7B          | untied | 256  | 0.000 | 0.000   | 0.003    | 0.010 | 0.000 |
| Granite-4.0 (Mamba) | hybrid | 256  | 0.003 | 0.000   | 0.003    | 0.001 | 0.000 |

**Table 10.** Read-strength robustness: training-free recall@1 (face,  $N=50$ , encoder ceiling 0.821) as the residual gain sweeps 16–1024. Tied models are exact throughout; untied models are exact above a threshold and stay exact. The constant tolerates a wide range.

| Model               | 16    | 32    | 64    | 128   | 256   | 512   | 1024  |
|---------------------|-------|-------|-------|-------|-------|-------|-------|
| Qwen2.5-3B (tied)   | 0.821 | 0.821 | 0.821 | 0.821 | 0.821 | 0.821 | 0.821 |
| Mistral-7B (untied) | 0.022 | 0.060 | 0.307 | 0.768 | 0.821 | 0.821 | 0.821 |

**Capacity law across modalities.** Table 11 runs the perceptual capacity probe (compress  $M$  registered identities into  $k$  prototype slots, recognise a cross-condition query, 5 seeds) on all five modalities. The slot-compression factor  $\min(1, k/M)$  fits every modality once normalised by the one-slot-per-identity ceiling  $C(M)$ : the normalised error is  $\leq 0.012$  on the strong-encoder modalities and  $\leq 0.13$  on the weaker style and tone encoders. The ceiling  $C(M)$  itself is the encoder’s recall and varies accordingly, from near 1 on voice to 0.26–0.36 at  $M=32$  on style and tone. Recognition is capacity-limited and graceful in every modality; only the ceiling moves.

**Table 11.** Perceptual capacity law per modality.  $C(M)$  is the one-slot-per-identity ceiling (recall at  $k=M$ ), i.e. the encoder’s recall at  $M$ . “fit” is the mean  $|\text{recall}/C(M) - \min(1, k/M)|$  over the  $k < M$  cells: the slot-compression factor is universal; the ceiling is the encoder’s.

| Modality       | $C(2)$ | $C(8)$ | $C(32)$ | fit to $\min(1, k/M)$ |
|----------------|--------|--------|---------|-----------------------|
| Face (ArcFace) | 0.98   | 0.95   | 0.92    | 0.002                 |
| Voice (ECAPA)  | 1.00   | 1.00   | 0.99    | 0.001                 |
| Acoustic (AST) | 0.95   | 0.87   | 0.76    | 0.012                 |
| Style (CLIP)   | 0.76   | 0.46   | 0.26    | 0.070                 |
| Tone (wav2vec) | 0.94   | 0.67   | 0.36    | 0.125                 |

**Learned-metric baseline.** A natural question is whether *any* re-encoding of the same features beats raw cosine, even if our read does not. We fit a similarity on the same encoder features and the same identities (the 50/50 split, seed 42) and score it with the identical protocol. Two closed-form metrics suffice: regularised LDA and within-class whitening. Table 12 shows they land on raw cosine at small memory and *below* it past a few hundred identities. So there is no discrimination left to extract from these features: the encoder’s cosine is already the best ruler, which is why the memory’s inheriting it exactly is the ceiling, not a shortfall. A contrastive

linear head we also trained underperformed both closed-form metrics and is omitted as a weak instantiation.

**Table 12.** Learned-metric baseline, recall@1, same encoder and same split as AttMem. A learned metric (LDA on faces, whitening on style) matches AttMem’s small- $N$  lift over raw cosine; at large  $N$  raw cosine wins for all. Style AttMem here is the seed-42 single run (the multi-seed mean is 0.64 at  $N=5$ ); the comparison is within seed.

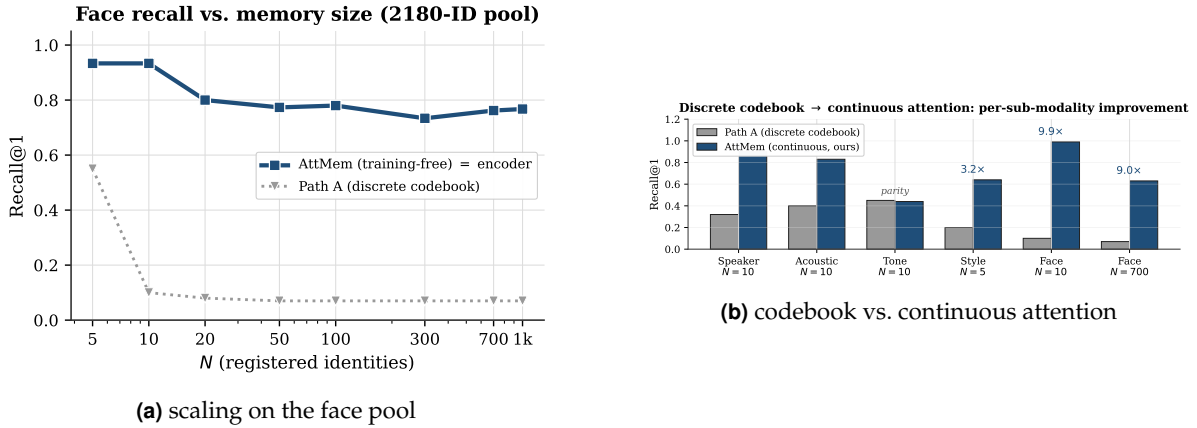
| Modality         | $N$  | Raw cosine | LDA          | Whitening    | AttMem       |
|------------------|------|------------|--------------|--------------|--------------|
| Face (ArcFace)   | 5    | 0.933      | <b>1.000</b> | 0.933        | 0.933        |
| Face (ArcFace)   | 10   | 0.933      | 0.967        | 0.933        | <b>1.000</b> |
| Face (ArcFace)   | 300  | 0.734      | 0.667        | <b>0.744</b> | 0.641        |
| Face (ArcFace)   | 1000 | 0.767      | 0.724        | <b>0.776</b> | —            |
| Style (CLIP-mid) | 5    | 0.400      | 0.200        | <b>0.600</b> | 0.467        |
| Style (CLIP-mid) | 10   | 0.400      | 0.333        | 0.367        | <b>0.467</b> |

**Comparison with per-concept methods.** The personalized-VLM line (MyVLM, Yo’LLaVA, Online-PVLM) registers a concept by *learning* something per concept (a classifier head, a soft token, a projection). Their code and checkpoints are unreleased, so we reimplement each method’s core mechanism charitably on the cross-condition face task (ArcFace keys, same split). Table 13 reports accuracy and registration cost. Two findings. First, a per-concept linear classifier (MyVLM’s mechanism) is statistically indistinguishable from raw cosine at every memory size, because with one registration embedding per identity the trained head collapses to the embedding itself. It buys no accuracy over retrieval, and it costs per-identity gradient descent that grows to 4.5 s at a thousand identities, against our  $\mathcal{O}(1)$  tensor append. Second, a learned projection trained with supervised contrastive loss (Online-PVLM’s mechanism) *underperforms* raw cosine by 15 to 28 points on this task. We read this as a caution that a learned re-encoding is not free: tuned on identity-disjoint data, it can distort the very cross-condition geometry it was meant to sharpen. No method here, ours included, beats raw cosine on accuracy at scale; the parametric memory’s case rests on cost and in-model behaviour, not recall.

**Table 13.** Charitable reimplementations of per-concept methods on cross-condition face recall (ArcFace, recall@1, single eval draw). MyVLM’s per-concept classifier ties raw cosine but its registration cost grows with the memory; Online-PVLM’s learned projection underperforms raw cosine. AttMem leads only at small  $N$  and trails at scale (the encoder-ceiling inversion).

| $N$  | Raw cosine | MyVLM | Online-PVLM | AttMem | MyVLM reg. cost |
|------|------------|-------|-------------|--------|-----------------|
| 5    | 0.933      | 0.933 | 0.800       | 0.933  | 0.009 s         |
| 10   | 1.000      | 0.967 | 0.750       | 0.992  | 0.019 s         |
| 50   | 0.767      | 0.753 | 0.675       | 0.733  | 0.178 s         |
| 100  | 0.780      | 0.770 | —           | 0.742  | 0.528 s         |
| 300  | 0.728      | 0.725 | —           | 0.637  | 1.39 s          |
| 1000 | 0.776      | 0.778 | —           | 0.594  | 4.52 s          |

**Scaling and the codebook comparison.** Figure 7a traces recall against memory size on the face pool. The training-free memory tracks the encoder’s own curve (it *is* that curve), declining gracefully from near-ceiling at ten identities to 0.77 at a thousand, while the discrete-codebook predecessor (PATH A) flatlines at  $\sim 0.07$  regardless of codebook size or budget. Figure 7b makes the same point across all five sub-modalities: replacing a discrete codebook with continuous attention lifts recall 2–10 $\times$ , because a categorical bottleneck discards the signal the encoder preserved. This is the one place a design choice in the read matters; everything else inherits the encoder.



**Figure 7.** Left: the continuous memory degrades gracefully with memory size (tracking the encoder ceiling), while the codebook predecessor never gets off the floor. Right: continuous attention beats the discrete codebook 2–10 $\times$  on every sub-modality.

**Vision-language model, key/value orthogonality.** Table 14 gives the numbers behind the structural rule of Section 6. Each memory row is compared to retrieval *in its own key space*. An external ArcFace key (orthogonal to the model’s hidden space) reproduces the win. The VLM’s native vision tokens (already in hidden space) only tie native-token retrieval and stay far below the ArcFace ceiling throughout, so they are the wrong key regardless of method.

**Table 14.** Parametric memory inside Qwen2.5-VL-3B: same frozen model, only the key encoder changes. Recall at three memory sizes.

| Configuration                                                | N=10         | N=100 | N=1000 |
|--------------------------------------------------------------|--------------|-------|--------|
| Retrieval over ArcFace embeddings (encoder ceiling)          | 0.933        | 0.780 | 0.767  |
| AttMem + external ArcFace keys (orthogonal to hidden space)  | <b>1.000</b> | 0.710 | 0.494  |
| Retrieval over native vision tokens                          | 0.40         | 0.35  | —      |
| AttMem + native vision tokens (co-located with hidden space) | 0.40         | 0.11  | —      |

## B Anatomy of a single recall

It helps to watch one recall end to end. Suppose the agent has met ten people and wants to register an eleventh from a single photo. Registration is one row appended to the face bank (Listing 1). The encoder turns the photo into a 512-dimensional ArcFace embedding, the *key*.

The model’s own embedding for a freshly assigned marker token becomes the *value*, a vector of the model’s hidden width (2048 for Qwen2.5-3B). There is no gradient step and no fine-tuning. The row is the memory.

```
# Register identity #11 from a single photo -- no training:
k = l2_normalize(arcface(photo))      # key:   R^512 (encoder space)
m = assign_marker_token()             # e.g. "<id_11>"
v = model.input_embedding[m]          # value: R^2048 (model's own space)
bank.K = torch.cat([bank.K, k[None]]) # 0(1) append
bank.V = torch.cat([bank.V, v[None]])
```

**Listing 1.** One registered identity is one row. The key is the encoder’s view of the perception; the value is the model’s own vector for the marker token that names it. Adding the row is a single tensor append.

At recall, a new photo of one of the eleven arrives under different lighting and a year older. Its ArcFace embedding becomes the query  $q$ . Just before its output head, the model scores  $q$  against all eleven keys, softmaxes into attention weights  $w$ , and pulls back the weighted blend of values  $r = w^\top V$ . Because each value is the model’s own vector for a marker token, adding  $g W_o r$  to the hidden state lands as a clean boost on the right marker’s logit. The model’s next token is the remembered identity. The whole step is a few matrix multiplies dwarfed by the model’s forward pass.

The 0.98-vs-0.46 diagonal gap of Section 5 (Figure 5) is measured on exactly this probe: a held-out ten-identity bank, comparing the sharp attention weights against the raw encoder cosine they are computed from. The attention concentrates the cosine onto its top match, which is why the read recovers the encoder’s nearest neighbour as a single decisive marker rather than a diffuse blend.

### C The discrete-codebook predecessor

Before continuous attention we spent sixteen development cycles on PATH A: a discrete codebook that snaps each perception to one of  $K$  learned codes and remembers the code. It is the natural “compress the perception into a symbol” design, and it is a learned cousin of captioning, which is why its failure is instructive rather than embarrassing.

No setting broke it. Across codebook sizes  $K \in \{128, 256, 512, 1024\}$ , even after 100,000 steps of continual pretraining on the expanded face pool, top-1 recall at 300 identities stays pinned between 0.057 and 0.070 while embedding retrieval over the same encoder reaches 0.73. That is a roughly  $10\times$  gap that does not move with  $K$  (Table 15). Swapping ArcFace R50 for AntelopeV2 R100 trained on 360K identities did not help either.

**Table 15.** PATH A after 100K-step continual pretraining on the 2180-identity face pool, at 300 registered identities. Recall is pinned near 0.07 regardless of codebook size, even when the gate routes to the correct code about half the time.

| Codebook size $K$                  | 128   | 256   | 512   | 1024  |
|------------------------------------|-------|-------|-------|-------|
| PATH A recall@1                    | 0.057 | 0.064 | 0.070 | 0.070 |
| gate routes to correct code        | 0.43  | 0.51  | 0.42  | 0.54  |
| embedding retrieval (same encoder) | 0.73  |       |       |       |



The reason is a vice the codebook cannot escape, and the diagnostics name it exactly. Two pressures pull in opposite directions as  $K$  grows. A *small* codebook packs many people into each cell: at  $K=16$ , distinct identities collide in the same code 8.7% of the time, and once two people share a code they are indistinguishable forever after. A *large* codebook fixes that, with inter-identity collisions falling to 2.4% at  $K=128$ , but it shatters each identity across cells. The rate at which two cross-condition photos of the *same* person land in the same code drops from 0.33 to 0.20, so the query no longer routes to where the registration was stored. Net recall is squeezed from both sides and never escapes  $\sim 0.07$ . Even when the gate does route a query to the right code (about half the time, Table 15), every other identity sharing that cell is an equally good answer, so the final pick is near-random.

This is the same information loss as captioning, learned instead of written: any categorical bottleneck, whether a word or a code, throws away the continuous signal the encoder worked to preserve, and no amount of compute downstream can recover it. Continuous attention over the raw embedding never quantizes, and never pays this tax. PATH A is in the paper because its failure is the cleanest possible argument for the design that replaced it.

## References

- Yuval Alaluf, Elad Richardson, Sergey Tulyakov, Kfir Aberman, and Daniel Cohen-Or. MyVLM: Personalizing VLMs for user-specific queries. In *European Conference on Computer Vision (ECCV)*, 2024.
- Ruichuan An, Sihan Yang, Ming Lu, Kai Zeng, Yulin Luo, Ying Chen, Jiajun Yong, Huanqia Liang, Qi Huang, and Shanghang Zhang. MC-LLaVA: Multi-concept personalized vision-language model. *arXiv preprint arXiv:2411.11706*, 2024.
- Alexei Baeovski, Henry Zhou, Abdelrahman Mohamed, and Michael Auli. wav2vec 2.0: A framework for self-supervised learning of speech representations. *Advances in Neural Information Processing Systems*, 2020.
- Huiyu Bai, Runze Wang, Zhuoyun Du, Yiyang Zhao, Fengji Zhang, Haoyu Chen, Xiaoyong Zhu, Bo Zheng, and Xuejiao Zhao. Online-PVLM: Advancing personalized VLMs with online concept learning. *arXiv preprint arXiv:2511.20056*, 2025.
- Prateek Chhikara, Dev Khant, Saket Aryan, Taranjeet Singh, and Deshraj Yadav. Mem0: Building production-ready AI agents with scalable long-term memory. *arXiv preprint arXiv:2504.19413*, 2025.
- Jiankang Deng, Jia Guo, Niannan Xue, and Stefanos Zafeiriou. ArcFace: Additive angular margin loss for deep face recognition. In *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.
- Brecht Desplanques, Jenthe Thienpondt, and Kris Demuynck. ECAPA-TDNN: Emphasized channel attention, propagation and aggregation in TDNN based speaker verification. In *INTERSPEECH*, 2020.
- Yuan Gong, Yu-An Chung, and James Glass. AST: Audio spectrogram transformer. In *INTERSPEECH*, 2021.

- Haoran Hao, Jiaming Han, Changsheng Li, Yu-Feng Li, and Xiangyu Yue. RAP: Retrieval-augmented personalization for multimodal large language models. In *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
- Gary B Huang, Marwan Mattar, Tamara Berg, and Eric Learned-Miller. Labeled faces in the wild: A database for studying face recognition in unconstrained environments. In *Workshop on Faces in Real-Life Images*, 2008.
- Urvashi Khandelwal, Omer Levy, Dan Jurafsky, Luke Zettlemoyer, and Mike Lewis. Generalization through memorization: Nearest neighbor language models. In *International Conference on Learning Representations (ICLR)*, 2020.
- Bojie Li. Models take notes at prefill: KV cache can be editable and composable. *arXiv preprint arXiv:2606.17107*, 2026a.
- Bojie Li. User as code: Executable memory for personalized agents. *arXiv preprint arXiv:2606.16707*, 2026b.
- Bojie Li. User as engram: Internalizing per-user memory as local parametric edits. *arXiv preprint arXiv:2606.19172*, 2026c.
- Bojie Li and Noah Shi. The latent bridge: A continuous slow-fast channel for real-time game agents. *arXiv preprint arXiv:2606.24470*, 2026.
- Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *International Conference on Machine Learning (ICML)*, 2023.
- Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Steven R Livingstone and Frank A Russo. The Ryerson audio-visual database of emotional speech and song (RAVDESS). volume 13, 2018.
- Lin Long, Yichen He, Wentao Ye, Yiyuan Pan, Yuan Lin, Hang Li, Junbo Zhao, and Wei Li. Seeing, listening, remembering, and reasoning: A multimodal agent with long-term memory (M3-Agent). *arXiv preprint arXiv:2508.09736*, 2025.
- Stylianios Moschoglou, Athanasios Papaioannou, Christos Sagonas, Jiankang Deng, Irene Kotsia, and Stefanos Zafeiriou. AgeDB: The first manually collected, in-the-wild age database. In *CVPR Workshops*, 2017.
- Thao Nguyen, Haotian Liu, Yuheng Li, Mu Yi, Jiarui Mu, and Yong Jae Lee. Yo’LLaVA: Your personalized language and vision assistant. *arXiv preprint arXiv:2406.09400*, 2024.
- Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G Patil, Ion Stoica, and Joseph E Gonzalez. MemGPT: Towards LLMs as operating systems. *arXiv preprint arXiv:2310.08560*, 2023.
- Vassil Panayotov, Guoguo Chen, Daniel Povey, and Sanjeev Khudanpur. LibriSpeech: An ASR corpus based on public domain audio books. 2015.

- Karol J Piczak. ESC: Dataset for environmental sound classification. In *ACM Multimedia*, 2015.
- Pine AI. Pine AI: The most natural human-computer interface is your voice. Blog post, 2026. URL <https://www.19pine.ai/blog/pine-ai-the-most-natural-human-computer-interface-is-your-voice>. Accessed 2026-06-28.
- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning (ICML)*, 2021.
- Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision. *arXiv preprint arXiv:2212.04356*, 2022.
- Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-Networks. In *Empirical Methods in Natural Language Processing (EMNLP)*, 2019.
- Letta Team. Letta (formerly MemGPT): Stateful LLM agents with long-term memory. *arXiv preprint arXiv:2310.08560*, 2024. Software framework.
- Yu Wang, Dmitry Krotov, Yuanzhe Hu, Yifan Gao, Wangchunshu Zhou, Julian McAuley, Dan Gutfreund, Rogerio Feris, and Zexue He. MemoryLLM: Towards self-updatable large language models. *arXiv preprint arXiv:2402.04624*, 2024.
- WikiArt. WikiArt: Visual art encyclopedia. 2010. <https://www.wikiart.org>.
- Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. Long-MemEval: Benchmarking chat assistants on long-term interactive memory. *arXiv preprint arXiv:2410.10813*, 2024.
- Yuhuai Wu, Markus N Rabe, DeLesley Hutchins, and Christian Szegedy. Memorizing transformers. In *International Conference on Learning Representations (ICLR)*, 2022.